



SOCIAL MEDIA ENGAGEMENT DASHBOARD

(Using Excel and Power BI)



1. PROJECT OVERVIEW

Social media platforms such as Instagram and YouTube generate a large amount of user interaction data every second. This includes likes, comments, shares, impressions, and views. Analyzing this data is essential for understanding audience behavior and improving content strategy.

This project develops a **Social Media Engagement Dashboard** using **Microsoft Power BI** to convert raw engagement data into meaningful insights. The dashboard provides visual analytics for better decision-making in digital marketing and content creation.



2. OBJECTIVE OF THE PROJECT

The main objectives of this project are:

- To understand the concept of social media engagement analytics
 - To analyze user interaction metrics such as likes, comments, and shares
 - To identify patterns in audience behavior
 - To evaluate content performance across different post types
 - To determine optimal posting times for higher engagement
 - To support data-driven decision-making in social media marketing
-



3. THEORETICAL BACKGROUND



3.1 Social Media Engagement

Social media engagement refers to the interaction between users and content posted on social media platforms. It is measured using metrics such as likes, comments, shares, saves, and

impressions. High engagement indicates that content is relevant, valuable, and appealing to the audience.

3.2 Importance of Engagement Analysis

Engagement analysis helps businesses and content creators:

- Understand audience preferences
 - Improve content quality
 - Increase brand visibility
 - Optimize posting strategy
 - Improve return on investment (ROI) in digital marketing
-

3.3 Key Engagement Metrics

- **Likes:** Measure of user appreciation
 - **Comments:** Indicates user interest and feedback
 - **Shares:** Shows content reach and virality
 - **Impressions:** Number of times content is displayed
 - **Engagement Rate:** Ratio of total interactions to total reach
-

3.4 Engagement Rate Formula

Engagement rate is a key performance indicator (KPI) used in social media analytics:

Engagement Rate = (Total Engagements / Total Followers) × 100

It helps measure how effectively content is engaging the audience.

3.5 Role of Data Visualization

Data visualization is the graphical representation of data using charts, graphs, and dashboards. It helps in:

- Simplifying complex data
- Identifying patterns quickly
- Supporting better decision-making
- Improving data interpretation

Power BI is widely used for this purpose due to its interactive and dynamic visualization features.

4. TOOLS & TECHNOLOGIES USED

- Microsoft Excel (Data cleaning and preprocessing)
 - Microsoft Power BI (Dashboard creation and visualization)
 - DAX (Data modeling and calculations)
 - Social media datasets (Instagram / YouTube analytics data)
-

5. DATA COLLECTION

Data for this project is collected from social media analytics sources such as:

- Instagram Insights
- YouTube Analytics
- Public datasets
- Simulated engagement datasets

Dataset includes:

- Post ID
- Post Type (Reel, Image, Video, Carousel)
- Date & Time of posting
- Likes
- Comments
- Shares
- Impressions
- Engagement Rate
- Category

6. METHODOLOGY (THEORETICAL + PRACTICAL)

The project follows a structured data analysis approach:

Step 1: Data Collection

Raw engagement data is collected from social media platforms or datasets.

Step 2: Data Cleaning

Data is cleaned in Excel by removing duplicates, handling missing values, and formatting columns.

Step 3: Data Transformation

Data is structured into a usable format for analysis in Power BI.

Step 4: Data Modeling

Relationships between different data fields are created in Power BI.

Step 5: Data Analysis

DAX formulas are used to calculate KPIs such as engagement rate and total likes.

Step 6: Data Visualization

Charts, graphs, and dashboards are created for better understanding.

Step 7: Insight Generation

Patterns and trends are analyzed to generate business insights.

7. DASHBOARD COMPONENTS

The dashboard includes the following components:

KPI Cards

Used to display key metrics such as:

- Total Likes

- Total Comments
 - Engagement Rate
 - Total Posts
-



Charts and Graphs

- Bar Chart → Comparison of post performance
 - Line Chart → Engagement trends over time
 - Column Chart → Post type analysis
-



Slicers (Filters)

Used for dynamic filtering of data:

- Date
 - Post Type
 - Category
-



8. ANALYSIS AND INTERPRETATION

The analysis of the dashboard reveals:

- Video-based content performs better than static posts
 - Engagement varies depending on posting time and content type
 - A small number of posts contribute to majority of engagement
 - Audience behavior changes over time
 - Consistent posting leads to higher engagement rates
-



9. POSTING TIME ANALYSIS

Time-based analysis shows that:

- Evening hours generally receive higher engagement
- Weekends show more user activity
- Posting consistency improves reach and visibility



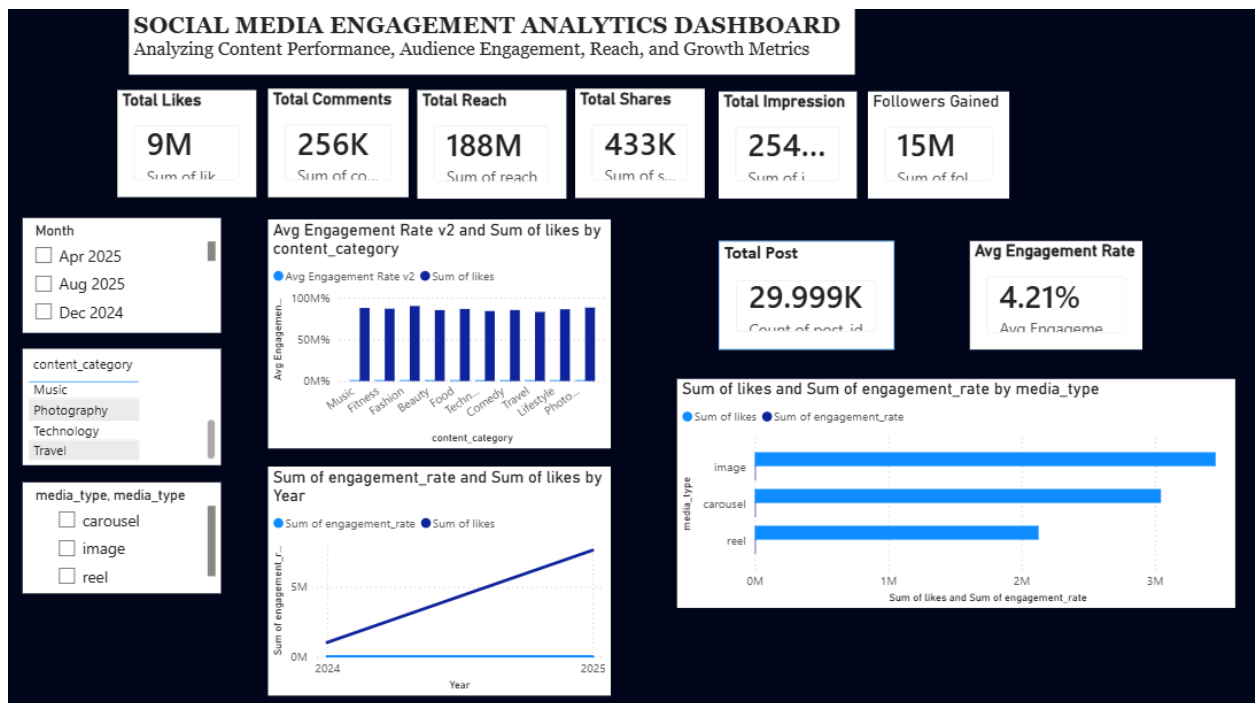
10. CONTENT STRATEGY INSIGHTS

Based on analysis, the following strategies are recommended:

- Focus on Reels and video content for higher engagement
- Post during peak activity hours
- Use trending hashtags and relevant topics
- Maintain consistent posting schedule
- Analyze past performance before creating new content



11. DASHBOARD VISUALIZATION



12. CONCLUSION

The Social Media Engagement Dashboard effectively demonstrates how raw social media data can be transformed into meaningful insights using Power BI. It helps in understanding user behavior, improving content strategy, and enhancing engagement performance.

This project highlights the importance of data analytics in modern digital marketing.



14. AUTHOR

Muskan

M.A. Economics Student | My Second Project |Data Analytics Enthusiast